poolside/Laguna-XS.2-FP8

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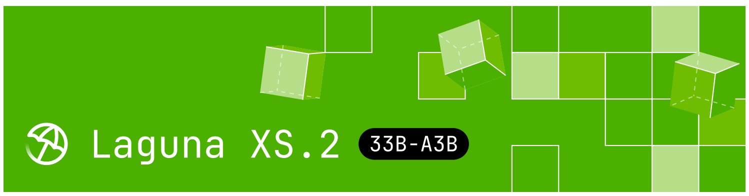


Try Laguna XS.2 in Shimmer · **Get an API key** · **Release blog post**

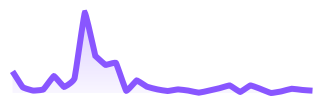
Laguna XS.2-FP8

Laguna XS.2-FP8 is a 33B total parameter Mixture-of-Experts model with 3B activated parameters per token designed for agentic coding and long-horizon work on a local machine. It uses Sliding Window Attention with per-head gating in 30 out of 40 layers for fast inference and low KV cache requirements.

> This is the FP8 variant with an FP8-quantized KV cache. The BF16, NVFP4 and INT4 variants are also available on Hugging Face.

Highlights

Downloads last month
1,977



Safetensors

Model size	33B params
Tensor type	F32 · BF16 · F8_E4M3

Chat template · Files info

Inference Providers NEW

Text Generation

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for poolside/Laguna-X...

Base model poolside/Laguna-XS.2
Quantized (15) .... this model

Collection including poolside/Lag...

Laguna XS.2 Collection
Designed for ... • 5 items • U... • 28

- **Mixed SWA and global attention layout**: Laguna XS.2 uses sigmoid gating with per-layer rotary scales, enabling mixed SWA (Sliding Window Attention) and global attention layers in a 3:1 ratio (across 40 total layers)
- **KV cache in FP8**: KV cache quantized to FP8, reducing memory per token
- **Native reasoning support**: Interleaved thinking between tool calls with support for enabling and disabling thinking per-request
- **Local-ready**: At 33B total parameters and 3B activated, Laguna XS.2 is compact enough to run on a Mac with 36 GB of RAM. Available on Ollama
- **Apache 2.0 license**: Use and modify freely for commercial and non-commercial purposes

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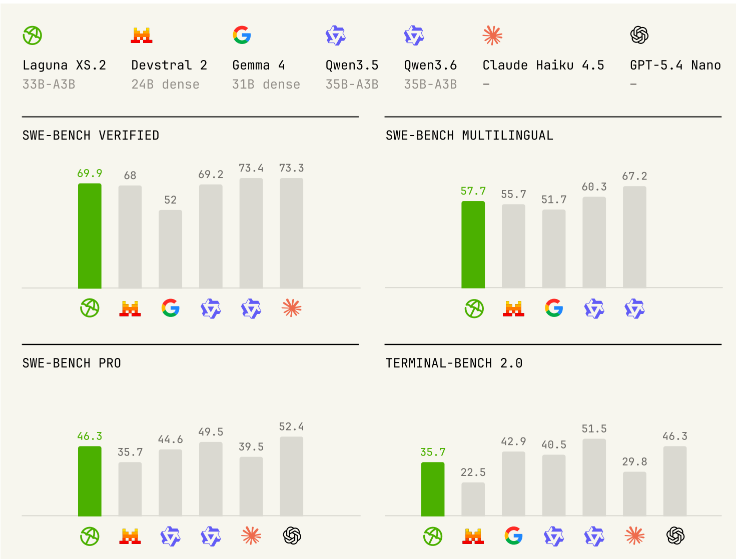
Model overview

- Training: pre-training, post-training and reinforcement learning stages
- Number of parameters: 33B total with 3B activated per token
- Optimizer: Muon
- Layers: 40 layers (10 layers with global attention, 30 layers with sliding window attention)
- Experts: 256 experts with 1 shared expert
- Sliding Window: 512 tokens
- Modality: text-to-text

- Context window: 262,144 tokens
- Reasoning support: interleaved thinking with preserved thinking

Benchmark results



Model	Size (total params.)	SWE-bench Verified	SWE-bench Multilingual	SWE-bench Pro (Public Datase
Laguna XS.2 (BF16)	33B	69.9%	57.7%	46.3%
Devstral Small 2	24B dense	68.0%	55.7%	-
Gemma 4 31B IT	31B dense	52.0%	51.7%	35.7%
Qwen3.5-35B-A3B	35B	69.2%	60.3%	44.6%
Qwen3.6-35B-A3B	35B	73.4%	67.2%	49.5%

Model	Size (total params.)	SWE-bench Verified	SWE-bench Multilingual	SWE-bench Pro (Public Datase
Claude Haiku 4.5	-	73.3%	-	39.5%
GPT-5.4 Nano	-	-	-	52.4%

We used the highest publicly-referenced scores for all comparison models across each benchmark. In almost all cases these were official scores published in release blog posts or equivalent, with the exception of Gemma 4 31B IT where the highest published scores were [reported by the Qwen team](#) and Claude Haiku 4.5 where the highest published (verified) scores for SWE-bench Pro and Terminal-Bench 2.0 are from their respective official leaderboards.

▶ Expand for benchmarking methodology

Usage

Laguna XS.2 has launch-day support in vLLM and Transformers, and TRT-LLM thanks to the support of the team at NVIDIA.

The fastest way to get started is with our API, directly or using OpenRouter.

> For complete usage instructions, see the main Laguna XS.2 model card.

Local deployment

Laguna XS.2 is supported in vLLM, SGLang, and Transformers, and TRT-LLM thanks to the support of the team at NVIDIA. Use Laguna-XS.2 with Ollama (with MLX support) and the mlx-lm framework for the best experience on your local machine.

vLLM

The full vLLM recipe is on the main Laguna XS.2 model card and on the vLLM recipes page. Quantization is detected automatically from `quantization_config` in this checkpoint, so the same command works with `poolside/Laguna-XS.2-FP8` substituted for the model ID. Set `VLLM_BLOCKSCALE_FP8_GEMM_FLASHINFER=0` when serving with vLLM.

> The FP8-quantized KV cache requires vLLM >= 0.22.0. Earlier versions produce scrambled output on non-Hopper GPUs because of a per-layer attention-head count bug, fixed in vllm#42650. On older vLLM, disable the FP8 KV cache by adding `--kv-cache-dtype-skip-layers $(seq 0 39)`.

SGLang

Laguna XS.2 is supported in SGLang via sgl-project/sglang#24204. Quantization is detected automatically from `quantization_config`, so no extra flags are required. See the SGLang cookbook entry and the main Laguna XS.2 model card for a serving recipe.

Transformers

The full Transformers recipe is on the main Laguna XS.2 model card. Substitute `poolside/Laguna-XS.2-`

FP8 for the model ID; quantization is detected automatically from quantization_config.

TRT-LLM

> Requires building TensorRT-LLM from the upstream PR that adds Laguna XS.2 support (NVIDIA/TensorRT-LLM#13559). Once that PR merges, the same code will work on a released tensorrt-llm wheel.

The full TRT-LLM recipe, including the laguna_minimal_overlay.sh step needed for transformers 4.57 compatibility, is on the main Laguna XS.2 model card. Quantization is detected automatically from quantization_config in this checkpoint, so no extra flags are required:

```
from tensorrt_llm import LLM

# OVERLAY built from poolside/Laguna-XS.2-FP
llm = LLM(model=OVERLAY, trust_remote_code=T
```

Ollama

Visit Ollama's model library to pull to your local machine.

Controlling reasoning

Laguna XS.2 has native reasoning support and is designed to work best with *preserved thinking*, where reasoning content from prior assistant messages is preserved in the message history. This model will generally reason before calling tools and between tool calls.

▶ Expand for example

Disabling reasoning

You can disable thinking by setting `enable_thinking` to `False` in a request or by not providing `--default-chat-template-kwargs {"enable_thinking": True}` or equivalent when starting the server.

▶ Expand for example

For agentic coding use cases, we recommend enabling thinking and preserving reasoning in message history as outlined in the [Controlling reasoning] section.

License

This model is licensed under the Apache 2.0 License.

Intended and Responsible Use

Laguna XS.2-FP8 is designed for software engineering and agentic coding use cases, and you are responsible for confirming that it is appropriate for your intended application. Laguna XS.2-FP8 is subject to the Apache 2.0 License, and should be used consistently with Poolside's Acceptable Use Policy. We advise against circumventing Laguna XS.2-FP8 safety guardrails without implementing substantially equivalent mitigations appropriate f

Please report security vulnerabilities or safety concerns to security@poolside.ai.